Zoo302 subjective

Fall 2022 final term by

MUHAMMAD HABIB

Molecular biology

Is the branch of biology that seeks to understand the molecular basis of biological activity in and between cells, including molecular synthesis, modification, mechanisms, and interactions. The study of chemical and physical structure of biological macromolecules is known as molecular biology.

Agonistic (Gr. Contest) behavior

, referring to any activity Related to fighting, whether it be aggression, defense, submission, or retreat.

Innate behavior

Behaviors that are closely controlled by genes with little or no environmental influence are called innate behaviors. These are behaviors that occur naturally in all members of a species whenever they are exposed to a certain stimulus. Innate behaviors do not have to be learned .

polymorphic locus .

Is one where two or more alleles exist.

 For example, imagine a researcher examined 20 loci from representatives of

Two populations.

Darwin also explained some of the differences that he saw.

In dryer regions, where vegetation was sparse, tortoises with longer necks

Would be favored because they could reach higher to get food.In moister regions, tortoises with longer necks would not necessarily be

Favored, and the shorter-necked tortoises could survive.

Geology

Is a branch of Earth science concerned with both the liquid and solid Earth, the rocks of which it is composed, and the processes by which they change over time. Geology can also include the study of the solid features of any terrestrial planet or natural satellite such as Mars or the Moon.

Vestigial Structures

Structures that have no apparent function in modern animals but clearly

Evolved from functioning structures in ancestors are called vestigial

Structures and provide another source of evidence of evolutionary changes

Polymorphism

Polymorphism occurs in a population when two or more distinct forms exist

Without a range of phenotypes between them.

Balanced polymorphism

Balanced polymorphism occurs when different

Phenotypes are maintained at relatively stable frequencies in the population and

May resemble a population in which disruptive selection operates.

Speciation

Is the formation of new species.

Allopatric speciation

Occurs when Subpopulations become geographically isolated from one another.

Sympatric speciation,

Occurs Within a single population.

Parapatric speciation,

Occurs in small, local populations, called demes.

Altruistic behaviour of animals

In evolutionary biology, an organism is said to behave altruistically when its behaviour benefits other organisms, at a cost to itself. The costs and benefits are measured in terms of reproductive fitness, or expected number of offspring.

Learning

Another aspect of behavior is learning, which we define as the modification

Of behavior through experience.

Adaptive traits

Traits that promote successful reproduction are said to be adaptive.

Adaptive traits become more common in subsequent generations.

Kin selection

Is the selection of genes by individuals assisting the

Survival and reproduction of individuals who possess the same genes byCommon descent.

Phylogeny

Refers to the evolutionary relationships among species.

 It includes the depiction of ancestral species and the relationships of modern

Descendants of a common ancestor.

Mutations:

changes in the structure of genes and chromosomes.

Mutations are the origin of all new alleles and a source of variation that may be adaptive for an animal.

Mutations make extinction less likely.

Gene therapy:

a field of manipulating human genes to treat or cure genetic diseases and

Disorders.

Polymorphic locus:

locus where two or more alleles exist.

Fossils:

evidence of plants and animals that existed in the past and have become incorporated Into the earth's crust (e.g., as rock or mineral).

Biotechnology:

use of biological techniques and engineered organisms to make products or

Plants and animals that have desired traits.

Biotechnology refers to the use of the biological sciences (such as gene manipulation), often

In combination with other sciences (such as materials sciences, nanotechnology, and Computer software), to discover, evaluate and develop products for bioindustry.

Means of Communication

i. Sounds
ii. Scents
iii. Touch
iv. Moveme

Fossils.

are formed in sedimentary rock by a variety of methods.

i. Sediments cover an organism to prevent scavenging and seal out oxygen and slows

Decomposition. Sediments continue to be piled on top of the dead organism, pressures

Build. Water infiltrates the remains and inorganic compounds and ions replace organic

Components. Hard parts of the organism are most likely to be fossilized, but delicate Structures are sometimes fossilized when silica is involved with replacement. These

Pressure and chemical changes transform the organism into a stony replica.

ii. Other fossilization processes form molds, casts of organisms, or carbon skeletons.
iii. Tracks and burrows, and even mummified remains, are sometimes found.

Effects of Gene Flow:

i. The exchange of alleles between an island population and a neighboring continental Population, for example, can change the genetic makeup of those populations.

ii. If gene flow continues and occurs in both directions, the two populations will become More similar.
iii. The absence of gene flow can make change in populations less likely.

Communication between Humans and Other Animals:

One uncertainty in studies of animal communication is understanding what sensory channel

An animal is using. The signals may be:

i. Visual displays
ii. Odors
iii. Vocalizations
iv. Tactile vibrations
v. Electrical currents (as, for example, among certain fishes).

Convergent Evolution:

the evolution of superficially similar structures in unrelated

Organisms is called convergent evolution, and the similar structures are said to be analogous.

Divergent evolution:

it shows how species can change slightly over time and separate (diverge) into new forms.

Genetic Drift: chance events influencing the frequencies of genes in populations result in

Genetic drift.

The chance of one or the other of two equally adaptive alleles being incorporated into a Gamete, and eventually into an individual in a second generation, is equal.

Two special cases promote genetic drift:

i. When a subset of a larger population colonizes a new habitat
ii. When a population undergoes a near extinction event.

Molecular biology.

Studying changes in anatomical structures and physiological processes reflects genetic Change and evolution.

Studying nuclear DNA, mitochondrial DNA, ribosomal RNA, and proteins is particularly Useful in evolutionary biology because these molecules can provide direct evidence of

Changes in genes and thus evolution.

RNA types.

i. Messenger RNA (mRNA): a linear strand that carries a set of genetic instructions for Synthesizing proteins to the cytoplasm.
ii. Transfer RNA (tRNA): it picks up amino acids in the cytoplasm, carries them to Ribosomes, and helps position them for incorporation into a polypeptide.
iii. Ribosomal RNA (rRNA): along with proteins, it makes up ribosomes.

Inherited variation sources

 Inherited variations exist.

 They arise from a variety of sources, including mutation, genetic Recombination and random fertilization.

Two forms of dance by Means honey bee communicate.

Waggle dance and round dance.

social aggregation in animals.

Many animals aggregate—which means they are socially attracted by others and group together—and receive in return benefits such as reduced predation risk or better access to resources or mates. When individuals aggregate, they interact

with other group members, which can change the behaviour of the group as a whole